

Turning Effect and Deformation Questions P2 5054

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0/N18/21/Q3

A student uses the apparatus shown in Fig. 3.1 to obtain the extension-load graph of a spring.

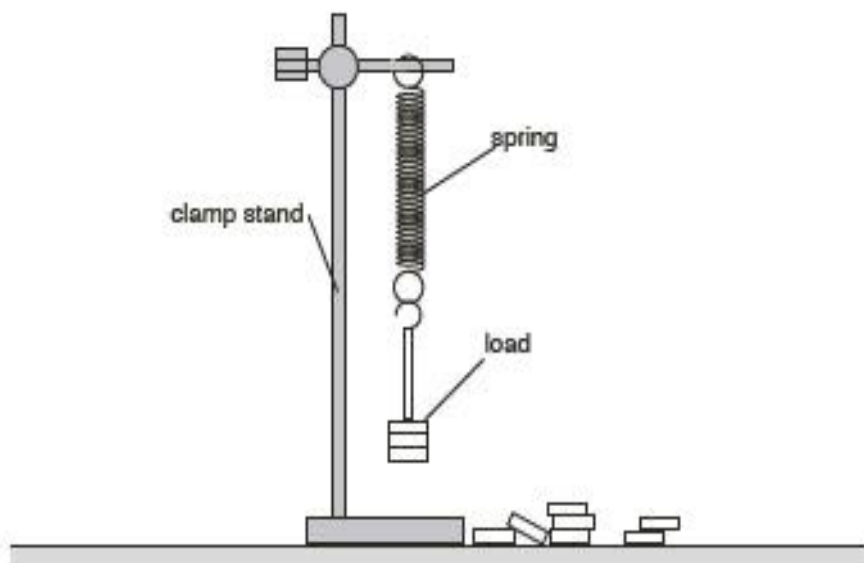


Fig. 3.1 (not to scale)

The weights of the loads are known. The student suspends loads of increasing size from the spring and determines the different extensions produced.

Fig. 3.2 is the graph the student obtains.

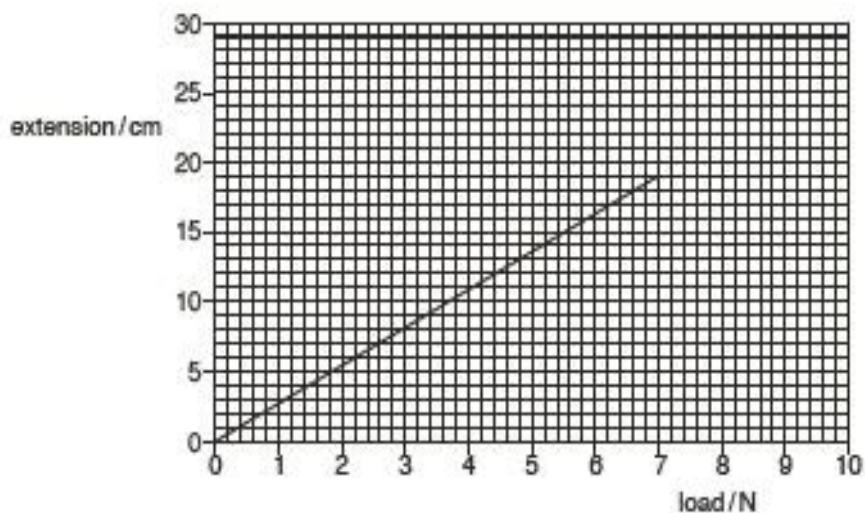


Fig. 3.2

The spring reaches its limit of proportionality when the load is 7.0 N.

(a) On Fig. 3.2, sketch the shape of the graph for loads greater than 7.0 N.

[1]

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- (b) All the loads are removed from the spring. A block of wood is suspended from the end of the spring. The length of the spring increases from 4.9 cm to 13.4 cm.

Determine the weight of the wooden block.

weight = [2]

- (c) The student takes a second spring that is identical to the one in Fig. 3.1 and hangs it next to the first spring.

Fig. 3.3 shows a light rod passing through the rings at the lower ends of the springs.

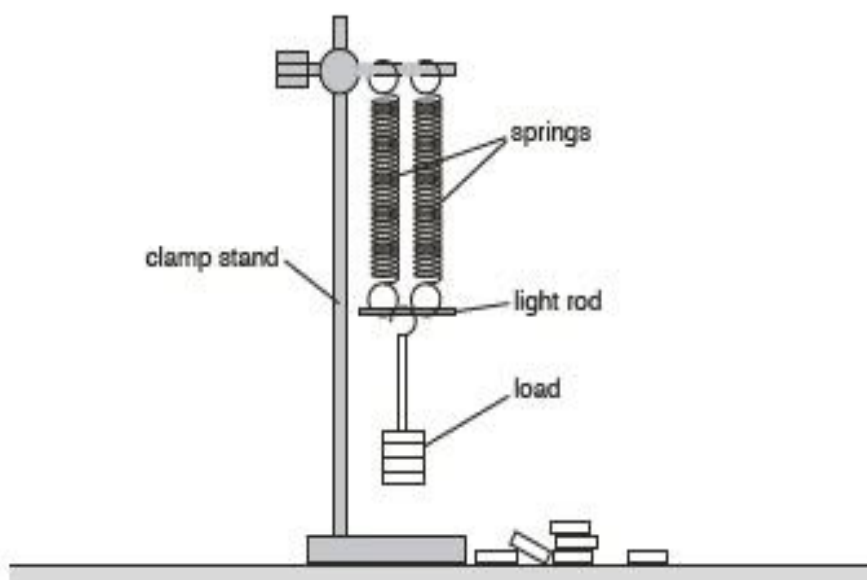


Fig. 3.3 (not to scale)

The student repeats the experiment and gradually increases the load to 9.0 N.

State and explain whether the limit of proportionality is reached.

.....
.....
.....
..... [2]

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M/J18/22/Q2

In an experiment, a student takes measurements and determines the extension of a spring for different loads. The apparatus is shown in Fig. 2.1.

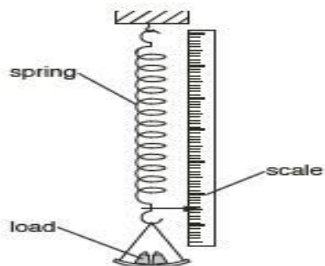


Fig. 2.1

The table in Fig. 2.2 shows the results.

load/N	0	1.0	2.0	3.0	4.0	5.0
length of spring/mm	200	235	270	305	350	420
extension/mm	0	35	70	105		

Fig. 2.2

- (a) Only some of the extensions are shown in the table.

Complete the table to show all of the extensions.

[1]

- (b) Calculate the load that produces an extension of 49 mm.

load =[2]

- (c) The student pulls a load downwards from position A to position B and holds it fixed at position B, as shown in Fig. 2.3.

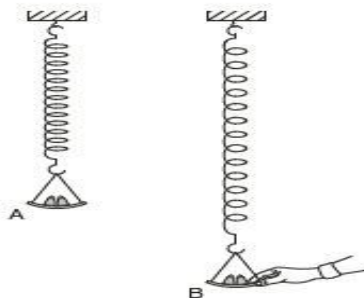


Fig. 2.3

The load is stationary at A and at B. The load has no kinetic energy at either point.

- (i) Place ticks (✓) in the boxes to show how the value of each of the forms of energy compares at A and B.

	equal at A and B	larger at A	larger at B
energy stored in spring	☐	☐	☐
gravitational potential energy of load	☐	☐	☐

[1]

- (ii) Work is done by the student to pull the load down. The law of conservation of energy states that energy cannot be created and cannot be destroyed.

Explain how this principle applies in this case.

.....

[2]

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0/N17/21/Q1

Fig. 1.1 shows a painter standing on a wooden plank, directly above the right-hand support.

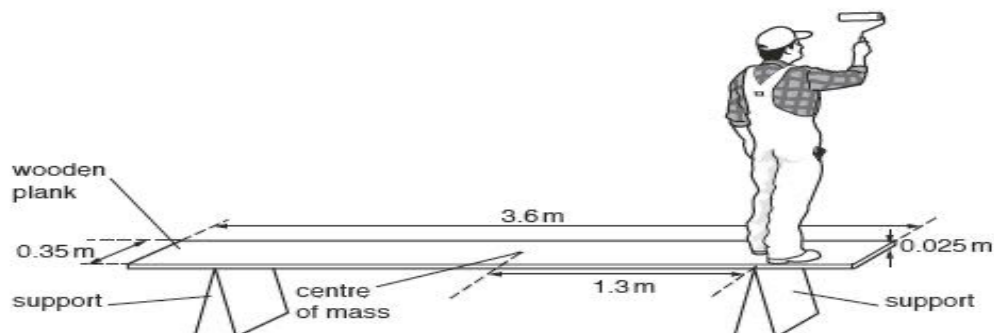


Fig. 1.1

The plank has length 3.6 m, width 0.35 m and thickness 0.025 m.

The gravitational field strength g is 10 N/kg and the mass of the plank is 23 kg.

(a) Calculate the density of the wood from which the plank is made.

density = [2]

(b) The centre of mass of the plank is in the middle of the plank at a distance of 1.3 m from each of the supports.

Calculate

(i) the weight of the plank,

weight = [1]

(ii) the moment of the plank about the right-hand support.

moment = [2]

(c) The painter moves further to the right along the plank and the plank rotates about the right-hand support.

Explain why the plank rotates.

.....
.....
.....
..... [1]

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0/N17/22/Q2

A force applied to a solid object may cause it to accelerate so that its velocity changes.

(a) State two other properties of the object which may change when a force is applied.

1.

2.

[2]

(b) A spring has a mass of 0.012 kg.

(i) The gravitational field strength g is 10 N/kg.

Calculate the weight of the spring.

weight = [1]

(ii) The spring is suspended vertically and a load is attached to its lower end. The spring extends by 2.7 cm and does not pass the limit of proportionality.

Fig. 2.1 shows the spring and load attached to the lower end of a second identical spring.

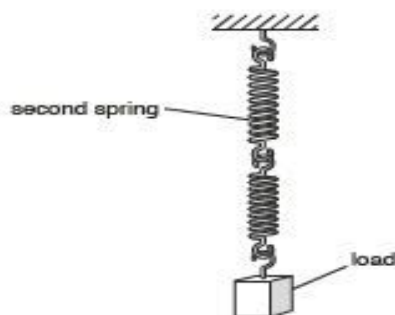


Fig. 2.1

Suggest one reason why the extension of the second spring differs from 2.7 cm.

.....
.....
..... [1]

(iii) Explain what is meant by *limit of proportionality*.

.....
.....
..... [1]

(c) The load in Fig. 2.1 is pulled down below its equilibrium position.

(i) State the form of energy stored in the stretched springs.

..... [1]

(ii) The load is released and it moves upwards and downwards. The distance travelled in each movement decreases until the load stops moving.

Explain why the load stops moving.

.....
.....
..... [1]

M/J17/22/Q3

Fig. 3.1 shows the brake pedal of a car which is connected to a brake cylinder.

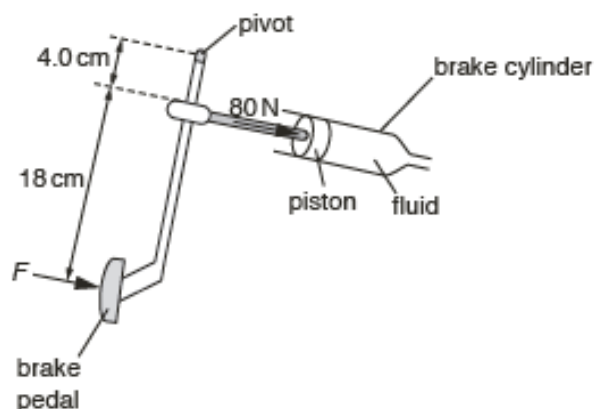


Fig. 3.1 (not to scale)

The brake is pressed with a force F . This force produces a moment about the pivot.

Pressing the brake causes a force of 80 N to act on the piston.

(a) Define the term *moment*.

.....
[2]

(b) Calculate the force F applied to the brake pedal.

$F =$ [2]

(c) The cross-sectional area of the piston is 0.0012 m^2 .

Calculate the pressure exerted by the brake piston on the fluid.

pressure =[2]

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0/N16/21/Q2

Fig. 2.1 shows a bed that folds away against a wall during the day.

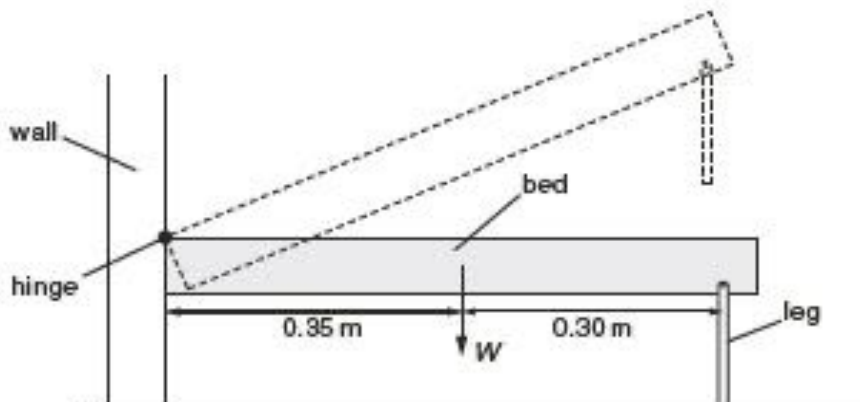


Fig. 2.1 (not to scale)

When it is horizontal, the bed is supported on one side by a hinge and on the other side by two legs. The weight W of the bed acts through its centre of mass, which is at a horizontal distance of 0.35 m from the hinge.

- (a) The mass of the bed is 28 kg. The gravitational field strength g is 10 N/kg.

Calculate the weight W of the bed.

$W =$ [1]

- (b) (i) State the principle of moments.

.....
.....
.....
.....[2]

- (ii) Determine the size of the total upward force exerted on the bed by the two legs when the bed is horizontal.

total force =[3]

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0/N16/22/Q2

When the lid of a freezer is opened, it pivots about the hinge at the back of the freezer. The handle is at the front.

Fig. 3.1 is a side view of the freezer.

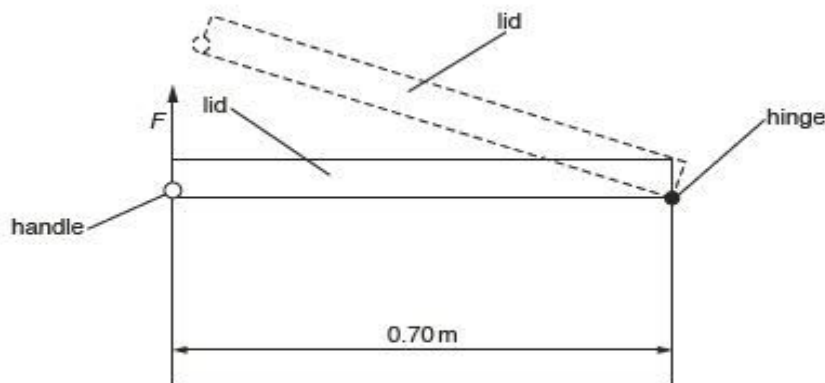


Fig. 3.1

The handle is 0.70 m from the hinge. The lid has a mass of 2.0 kg.

(a) The gravitational field strength g is 10 N/kg.

(i) Calculate the weight of the lid.

weight =[1]

(ii) The lid is uniform and its centre of mass is at its centre. The weight of the lid produces a moment about the hinge.

1. Calculate the moment about the hinge when the lid is closed.

moment =[2]

2. The moment required to open the lid is greater than the value calculated in (a)(ii)1.

Suggest one reason for this.

.....
.....[1]

(b) The lid is closed. To open the lid, a force F is applied to the handle as shown in Fig. 3.1.

The direction of F is vertically upwards and F is the smallest possible force that opens the lid.

A force on the handle in any other direction must be larger than F in order to open the lid. Explain why.

.....
.....
.....[1]

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O/N16/22/Q8

A spring is taken from a drawer in a physics laboratory.

(a) Fig. 8.1 is the extension-load graph for the spring.

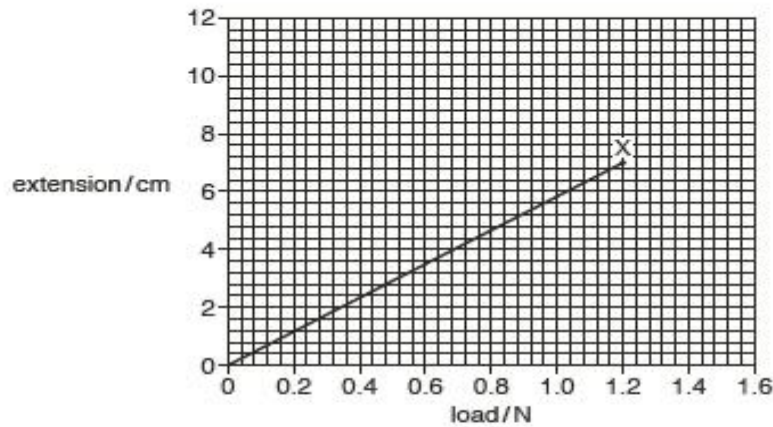


Fig. 8.1

(i) Using Fig. 8.1, determine the load on the spring when its extension is 4.9 cm.

load = [1]

(ii) The point X on Fig. 8.1 is the limit of proportionality for the spring.

On Fig. 8.1, sketch a line to show how the extension-load graph might continue after X. [2]

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- (b) A rectangular, plastic block is suspended from the lower end of the spring. Fig. 8.2 shows that the bottom 3.5 cm of the plastic block is in the water in a beaker.

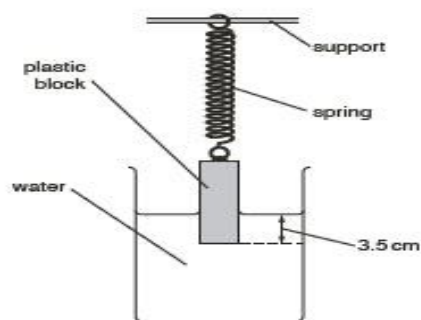


Fig. 8.2 (not to scale)

The density of water is 1000 kg/m^3 and the gravitational field strength g is 10 N/kg . The base of the plastic block is a square with sides of 4.0 cm .

- (i) Calculate the pressure due to the water at a depth of 3.5 cm below the surface of the water.

pressure = [3]

- (ii) Calculate the upward force on the base of the block due to the pressure of the water.

force = [2]

- (iii) In Fig. 8.2, the extension of the spring is 4.9 cm .

Using the answer from (a)(i) and the answer from (b)(ii), determine the weight of the block.

weight = [1]

- (c) In the equipment shown in Fig. 8.2, the pressure of the atmosphere is transmitted through the water. The total pressure in the water at a depth of 3.5 cm is larger than the pressure due to the water alone.

- (i) Explain why, in this experiment, atmospheric pressure does not affect the position of the plastic block in the water.

.....
.....
.....
..... [2]

- (ii) Force is a vector quantity and pressure is a scalar quantity.

State how a vector quantity differs from a scalar quantity.

.....
..... [1]

- (d) The beaker in Fig. 8.2 is gradually moved upwards so that more of the plastic block is in the water.

Explain what happens to the forces that act on the block.

.....
.....
.....
.....
.....
..... [3]

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M/J16/21/Q2

A student hangs various masses from the end of a spring and determines the extension produced. Fig. 2.1 shows the spring hanging vertically on its own and with an unknown mass X at one end.

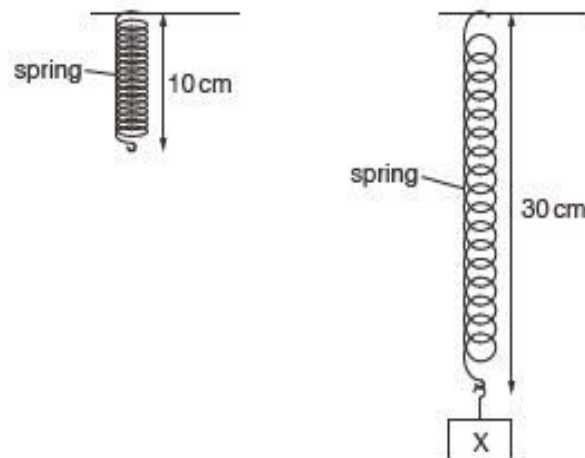


Fig. 2.1 (not to scale)

- (a) The student plots a graph of the extension of the spring against the mass hanging on the spring. Fig. 2.2 shows the student's graph.

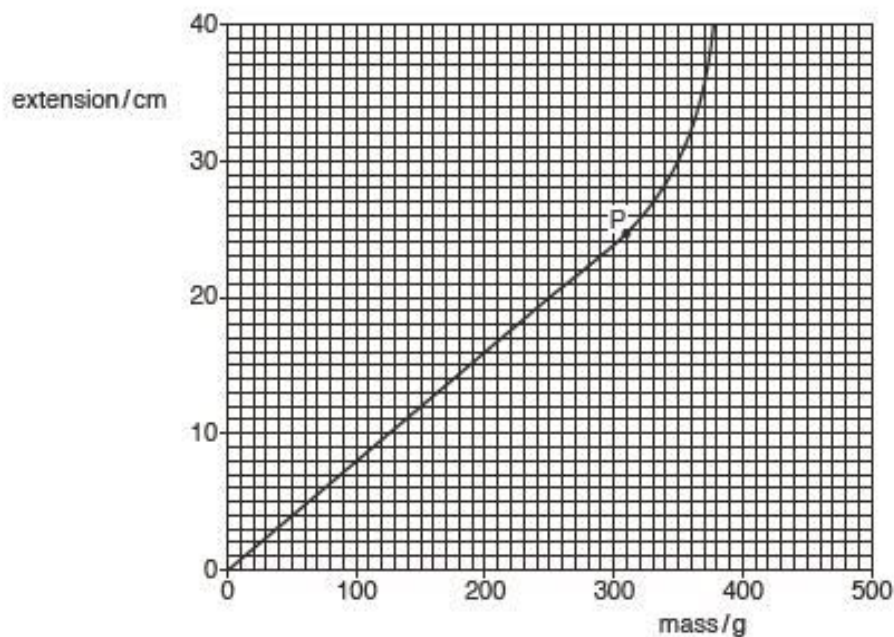


Fig. 2.2

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- (i) At point P on the graph, the line begins to curve.

State the name of point P.

.....[1]

- (ii) Using Fig. 2.1 and Fig. 2.2, determine the mass X.

mass = [1]

- (iii) The gravitational field strength g is 10 N/kg .

Calculate the weight of X.

weight = [1]

- b) An identical spring is used with the original spring, as shown in Fig. 2.3.

Together they support the mass X.

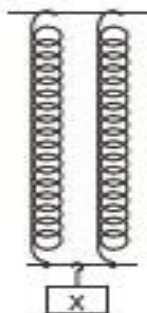


Fig. 2.3 (not to scale)

State and explain how the extension in Fig. 2.3 compares with the extension in Fig. 2.1.

.....
.....
.....
.....[2]

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M/J16/22/Q3

A student has three springs A, B and C. He measures the length of each spring, in turn, when different weights are placed on the end of each spring.

His table of results is shown in Fig. 3.1.

weight/N	length of spring A/cm	length of spring B/cm	length of spring C/cm
1.0	6.1	8.6	9.7
1.5	6.9	9.5	10.5
2.0	7.7	10.4	11.3
2.5	8.5	11.2	13.1
3.0	9.3	12.1	16.9

Fig. 3.1

(a) (i) State which spring has been stretched past the limit of proportionality.

.....

(ii) Using data from Fig. 3.1, explain how you obtained your answer to (a)(i).

.....

.....

.....

.....

[2]

(iii) Calculate the unstretched length of spring A.

unstretched length =[1]

(b) Describe how the student can use spring A to determine the mass of a small rock.

.....

.....

.....

.....

.....[2]

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0/N15/21/Q4

Fig. 4.1 shows a screwdriver of mass 64g resting in equilibrium on a pivot.

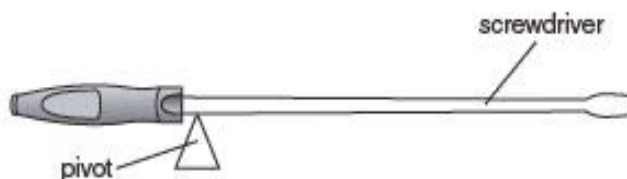


Fig. 4.1

(a) On Fig. 4.1, mark and label with a C, the centre of mass of the screwdriver. [1]

(b) The gravitational field strength is 10N/kg.

(i) Calculate the weight of the screwdriver.

weight = [1]

(ii) On Fig. 4.1, draw an arrow labelled *W* to represent the weight of the screwdriver. [1]

(c) State two conditions that apply when an object is in equilibrium.

1.
.....

2.
.....
.....

[2]

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0/N15/22/Q2

- (a) The surface of a running track is made of rubber. A heavy trolley is pulled on to the track and it exerts a large force on the rubber track.

State two effects that this force has on the rubber.

1.
2.

[2]

- (b) A spring is suspended from a support with a small pan attached to its lower end. Masses are added to the pan until the spring is extended well beyond the limit of proportionality.

Fig. 2.1 shows this apparatus.

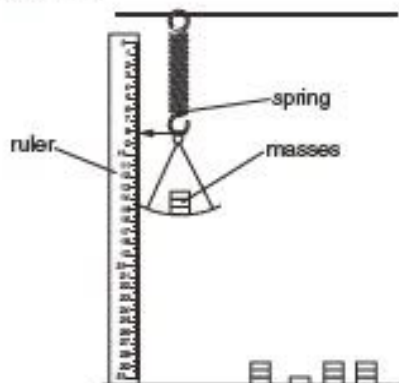


Fig. 2.1

A ruler is used when determining the extension of the spring.

- (i) On Fig. 2.2, sketch the extension-load graph for the spring and label the limit of proportionality P. [2]



Fig. 2.2

- (ii) The masses are then removed and the extension of the spring decreases.

Suggest what is observed when all the masses are removed.

.....
..... [1]

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M/J15/21/Q2

Fig. 2.1 shows a student doing a press-up. A total force F acts upwards on his hands. There is also a force upwards on his toes.

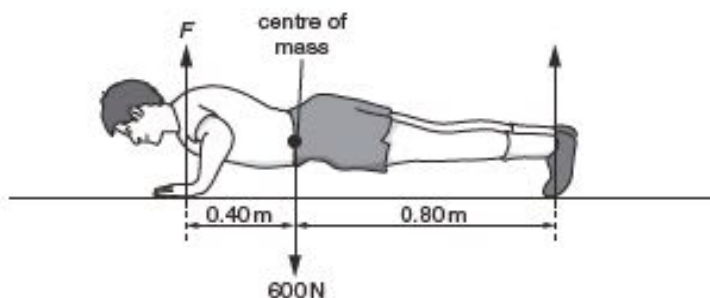


Fig. 2.1 (not to scale)

The weight of the student is 600 N and this force acts downwards from his centre of mass.

- (a) (I) Explain why the student does work as his body rises from the ground.

.....
.....[1]

- (II) State the form of energy that the student uses to do this work.

.....[1]

- (b) At the position shown in Fig. 2.1, the student is stationary.
The weight of the student causes a moment about his toes.

- (I) Calculate the moment of the weight of the student about his toes.

moment =[1]

- (II) Calculate the value of the force F .

F =[2]

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M/J15/22/Q1

The apparatus shown in Fig. 1.1 is used to measure the extension of a spring.

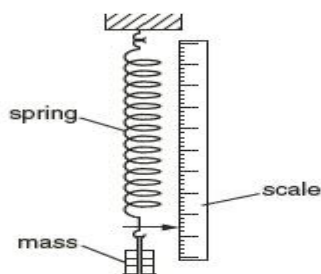


Fig. 1.1

- (a) Explain how the mass causes a force on the spring.

..... [1]

- (b) The force on the spring is a vector quantity. State what is meant by a *vector* quantity.

..... [1]

- (c) Fig. 1.2 shows a graph of the length of the spring plotted against the force on the spring, for forces between 0 and 10 N.

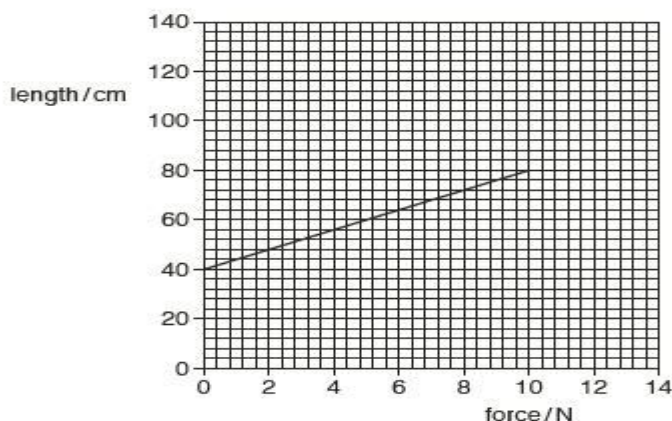


Fig. 1.2

- (i) 1. State a formula that relates the unstretched length l_0 of the spring, the stretched length l of the spring and the extension e of the spring.

..... [1]

2. A mass produces a force of 9.0 N on the spring. Determine the extension of the spring caused by this mass.

extension = [1]

- (ii) The limit of proportionality of the spring is reached when the force is 10 N.

The spring is easier to stretch after the limit of proportionality.

On Fig. 1.2, continue the line to suggest how the length changes when the force is greater than 10 N. [1]

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0/N14/21/Q2

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A student places a metre rule on the edge of a triangular prism. The prism is used as a pivot and the rule balances about the 50cm mark.

The student then places a block of wood at the 10cm mark on the rule and an empty measuring cylinder at the 80cm mark. The rule is still balanced. Fig. 2.1 shows the arrangement.

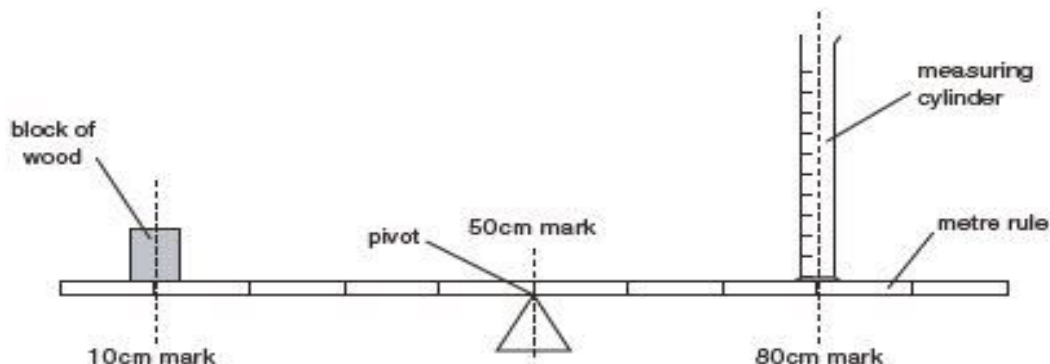


Fig. 2.1 (not to scale)

The student now places a weight of 0.39 N on top of the block of wood. She then starts to pour oil into the measuring cylinder. The rule balances again when there is 60 cm³ of oil in the measuring cylinder.

The gravitational field strength g is 10 N/kg.

(a) Calculate

(i) the weight of the oil in the cylinder,

weight = [2]

(ii) the mass of the oil in the cylinder.

mass = [1]

(b) Calculate the density of the oil.

density = [2]

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0/N14/22/Q1

A student suspends a spring from a clamp stand. He measures the unstretched length of the spring with no load attached. He then attaches different loads to the lower end of the spring and measures the new lengths of the spring.

Fig. 1.1 shows how the total length of the spring depends on the load attached.

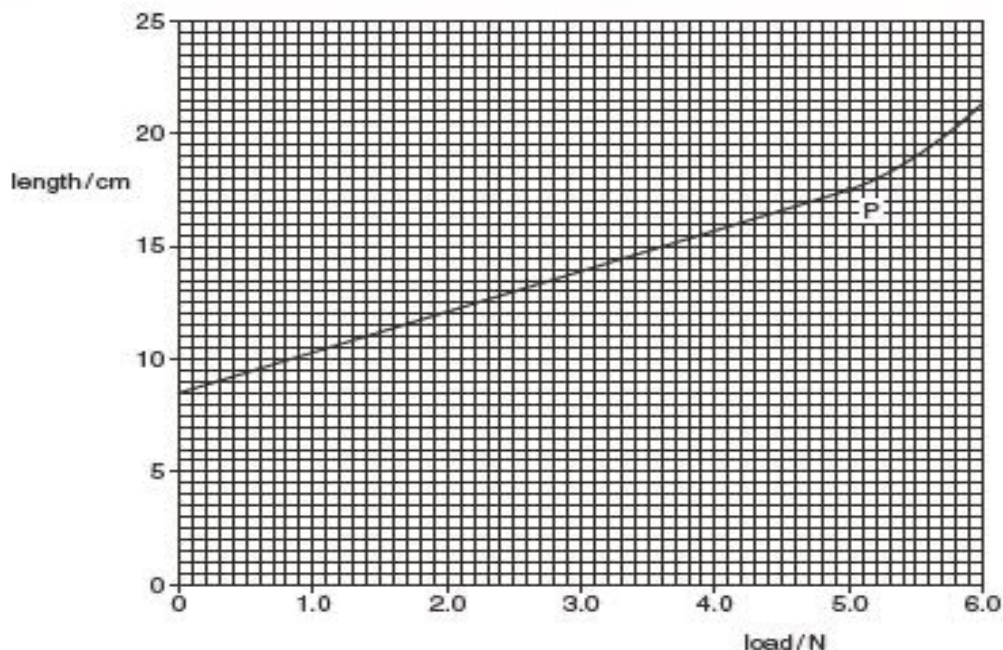


Fig. 1.1

- (a) At point P on Fig. 1.1, the line stops being straight and begins to curve.

State the name of point P.

.....[1]

- (b) (i) Using Fig. 1.1, determine the unstretched length of the spring.

unstretched length =[1]

- (ii) Calculate the extension of the spring for a load of 4.0 N.

..... extension =[1]

- (c) The student attaches a small block of wood to an identical spring. The extension of the spring is 2.7 cm. The gravitational field strength g is 10 N/kg.

Calculate the mass of the block of wood.

mass =[2]

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0/N14/22/Q2

In hospitals, doctors and nurses operate taps with their elbows in order to avoid contamination.

Fig. 2.1 shows a hospital tap with a long handle.

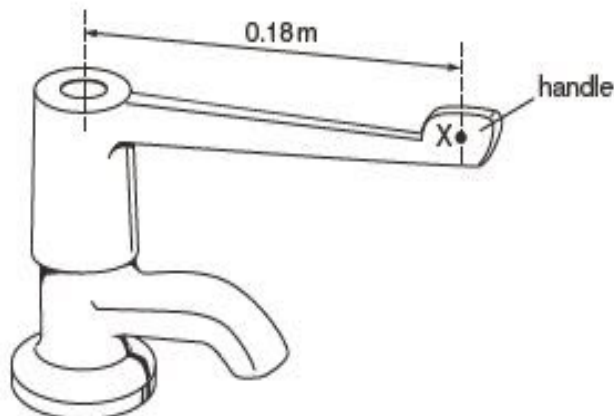


Fig. 2.1

(a) A nurse applies a force of 2.5 N at a point X on the handle, 0.18 m from the axis of the tap.

(i) Calculate the maximum moment about the axis that this force can produce.

moment =[2]

(ii) The moment produced by the nurse is less than this maximum value.

Suggest one reason why this is so.

.....
.....[1]

(b) State how the force needed to operate the tap is affected by the length of the handle.

.....
.....[1]

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0/N13/21/Q2

A girl of weight 550 N is playing on a see-saw with her brother. Fig. 2.1 shows her brother of weight W sitting 1.1 m to the right of the balance point.

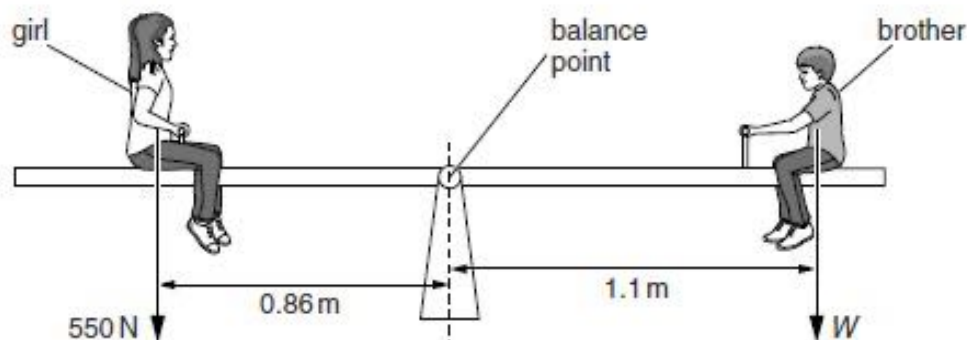


Fig. 2.1 (not to scale)

The see-saw is balanced when the girl sits 0.86 m to the left of the balance point.

(a) Calculate W .

$W = \dots\dots\dots$ [2]

(b) The girl and her brother slide equal distances along the see-saw away from each other.

Describe and explain what happens.

.....
.....
.....
..... [3]

0/N13/22/Q9

- (a) State what is meant by the *moment of a force*.

.....
.....[2]

- (b) The anchor of a sailing ship has a mass of 350 kg. Six sailors raise the anchor from the sea-bed by turning a large axle. They push the handles attached to the axle and it rotates. The anchor is on the end of a chain that winds on to the rotating axle.

Fig. 9.1 shows the sailors lifting the anchor.

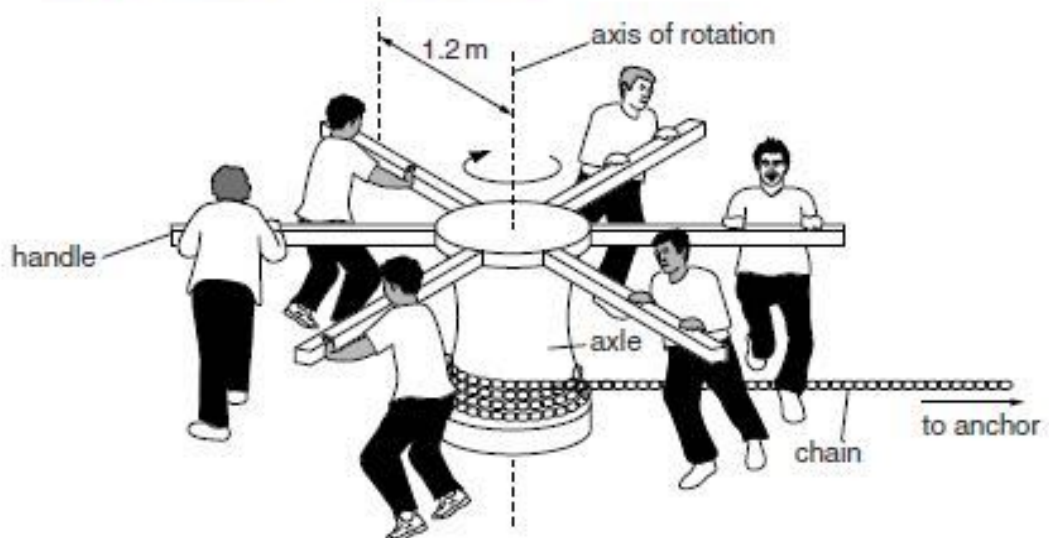


Fig. 9.1

Each of the sailors exerts a force of 750 N on his handle at a distance of 1.2 m from the axis of rotation. The axle rotates through one complete revolution and the anchor is lifted through a distance of 160 cm.

The gravitational field strength g is 10 N/kg.

- (i) Calculate

1. the total moment exerted on the axle by the six sailors,

moment =[2]

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2. the gravitational potential energy gained by the anchor as the axle rotates through one complete revolution.

energy = [3]

- (ii) The work done on the axle by the sailors is very much larger than the gravitational potential energy gained by the anchor.

State and explain how energy is wasted.

.....
.....
..... [2]

- (iii) Explain why the power produced by the sailors is larger when the anchor is lifted at a faster rate.

.....
..... [2]

- (c) Describe, with the aid of a labelled diagram, how to verify the principle of moments.

.....
.....
.....
.....
.....
..... [4]

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M/J13/21/Q1

Fig. 1.1 shows how the length of a spring varies as the force applied to it increases.

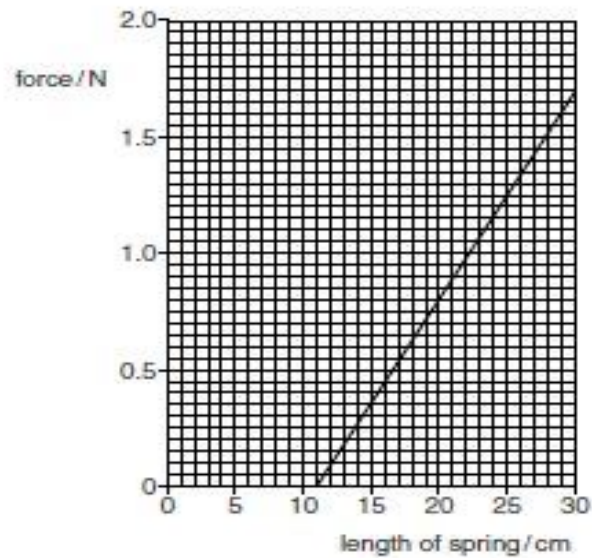


Fig. 1.1

- (a) Determine the length of the unstretched spring.

length = [1]

- (b) Explain how the graph shows that the limit of proportionality is not reached.

.....
..... [1]

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- (c) The spring is attached to a mass M of 0.20 kg and placed on a frictionless surface, as shown in Fig. 1.2.

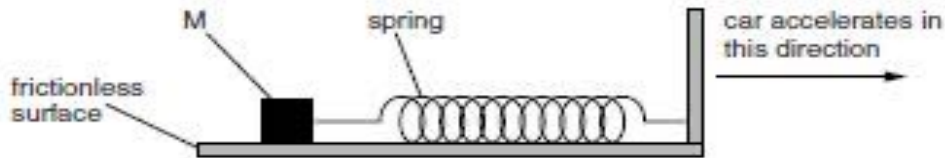


Fig. 1.2 (not to scale)

The apparatus is placed on the floor of a car.

When the car accelerates uniformly in the direction shown, the spring extends.

- (I) State what is meant by a *uniform acceleration*.

.....
.....
..... [2]

- (II) The extension of the spring is 9.0 cm .

Using Fig. 1.1, determine

1. the horizontal force on M ,

force = [1]

2. the acceleration of M .

acceleration = [2]

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M/J13/21/Q3

Fig. 2.1 shows apparatus used to investigate the turning effect of a force.

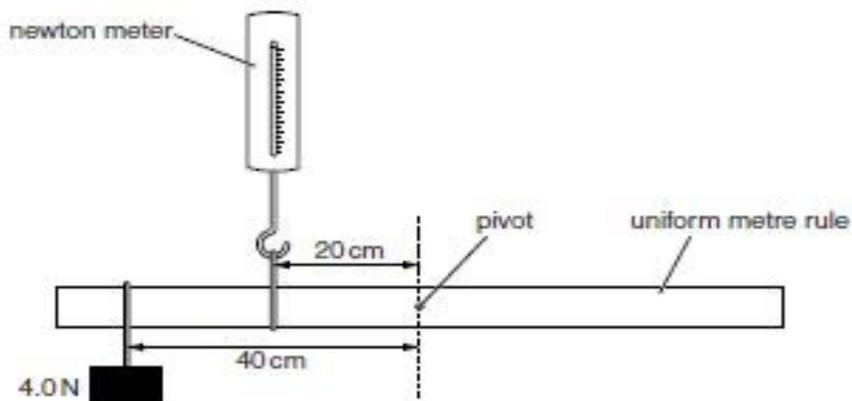


Fig. 2.1

The uniform metre rule is freely pivoted at its centre.

The newton meter is 20 cm from the pivot and a 4.0 N weight is 40 cm from the pivot.

The metre rule is in equilibrium.

(a) State the principle of moments for a body in equilibrium.

.....
.....
..... [1]

(b) Calculate the reading on the newton meter.

reading = [2]

(c) The weight of the metre rule is 1.2 N.

Calculate the size of the force exerted on the metre rule by the pivot.

force = [2]

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M/J13/22/Q2

- (a) Explain what is meant by the *moment of a force*.

.....

.....

..... [2]

- (b) Fig. 2.1 shows a system for raising a heavy piece of metal into a vertical position.

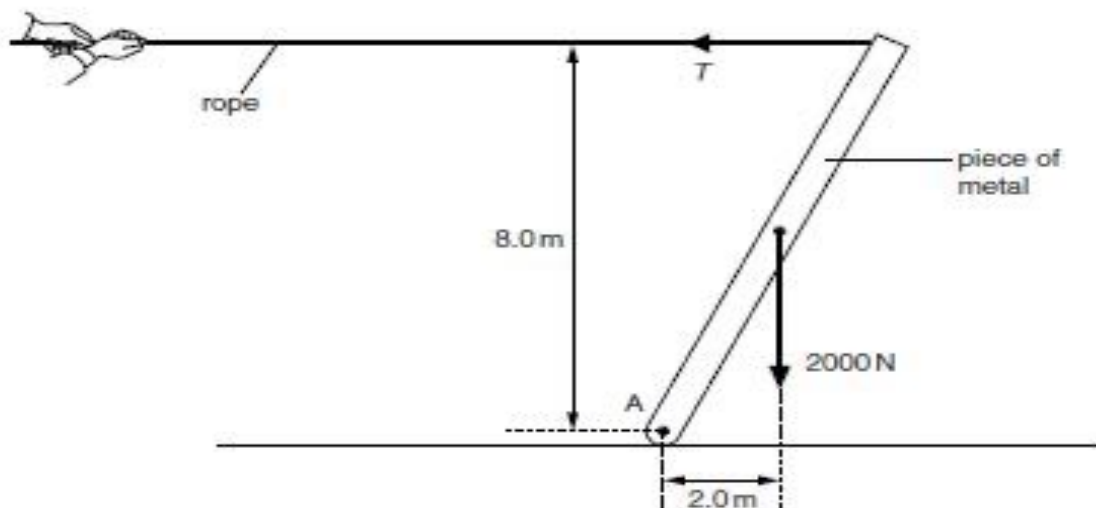


Fig. 2.1 (not to scale)

A man pulls on the rope with a horizontal force T . The piece of metal has a weight of 2000 N and is freely pivoted at A. The system is in equilibrium.

- (i) By taking moments about A, calculate T .

$T = \dots\dots\dots$ [2]

- (ii) The force T and the force that the rope exerts on the man are related by Newton's third law. State the relationship between these forces.

.....

.....

..... [2]

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M/J13/22/Q3

Fig. 3.1 shows a man using a chest expander to increase the strength of his arms.



Fig. 3.1

The chest expander may be considered to be a single spring.

Fig. 3.2 shows the force needed to produce different extensions of the spring.

force / N	extension / cm
0	0
60	10
120	20
180	30

Fig. 3.2

- (a) The man increases the force on the spring from 0 to 180 N.
The spring extends by 30 cm and the average force exerted during this process is 90 N.

(i) Calculate the work done on the spring.

work = [2]

(ii) Twenty extensions are made in 1.0 minute.

Calculate the power used to extend the spring.

power = [2]

- (b) The force at the limit of proportionality of the spring is 800 N.

(i) Calculate the extension of the spring when the force is 800 N.

extension = [2]

(ii) The force on the spring is increased from 800 N to 860 N.

Suggest what happens to the extension.

.....
..... [1]

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0/N12/22/Q1

- (a) Describe an experiment to verify the *principle of moments*. You may include a diagram in your answer.

.....

.....

.....

.....

..... [4]

- (b) Fig. 1.1 shows a spanner tightening a nut.

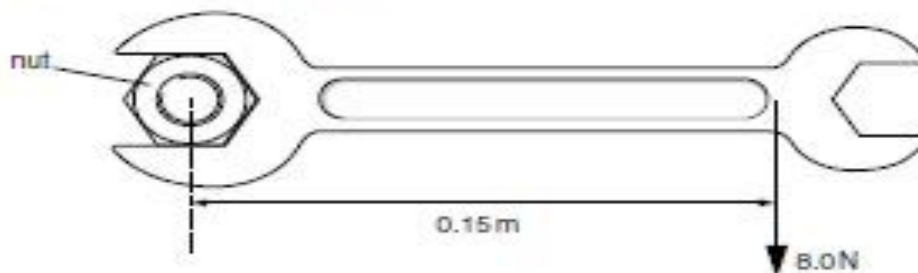


Fig. 1.1

A force of 8.0 N is applied to the spanner at a perpendicular distance of 0.15 m from the centre of the nut.

Calculate the moment of the force acting on the nut.

moment = [2]

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0/N12/22/Q2

A pupil hangs a load of 45 N from a length of copper wire. The wire extends by 2.5 mm and does not exceed the limit of proportionality.

- (a) (I) The gravitational field strength is 10 N/kg. Calculate the mass of the 45 N load.

mass = [1]

- (II) Use the grid in Fig. 2.1 to plot the extension-load graph for this wire for a load between 0 and 45 N.

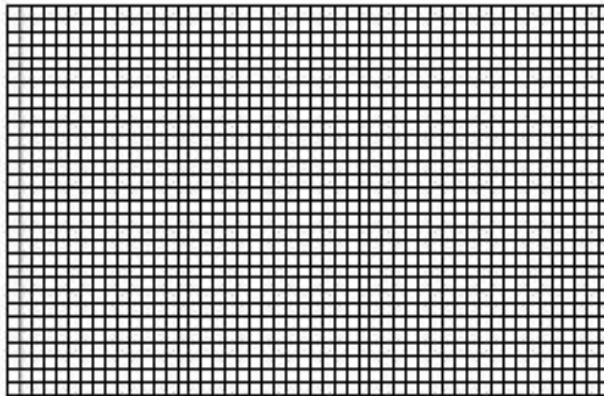


Fig. 2.1

[3]

- (b) Use the graph plotted in Fig. 2.1 to determine the load needed to produce an extension of 1.3 mm.

load = [1]

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M/J12/22/Q1

Fig. 1.1 shows apparatus used to obtain the readings for a graph of force against extension for a spring.

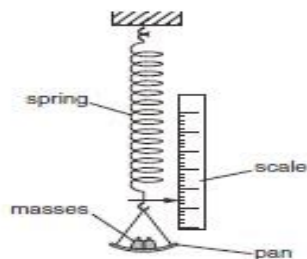


Fig. 1.1

The masses added to the pan produce a force that stretches the spring.

- (a) (i) State what is meant by the *mass* of a body.

.....
.....[1]

- (ii) Describe how the scale is used to find the extension of the spring.

.....
.....
.....[1]

- (b) Fig. 1.2 shows the force-extension graphs for two different springs.

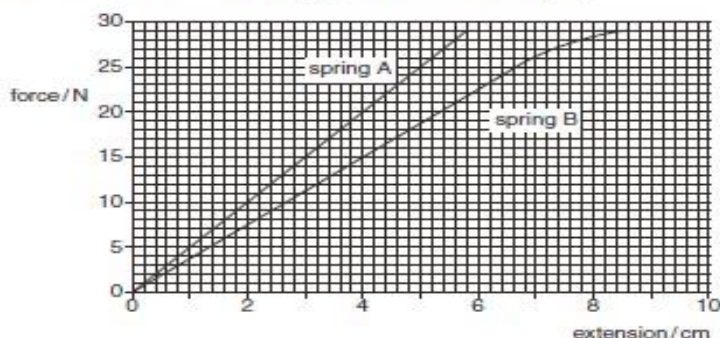


Fig. 1.2

- (i) A student states that spring B is easier to stretch than spring A.

Use values from Fig. 1.2 to explain what the student means.

.....
.....
.....[1]

- (ii) When a force of 25 N is applied, spring B reaches its limit of proportionality but spring A does not. Explain how Fig. 1.2 shows this.

.....
.....
.....[1]

- (iii) The same force is applied to each spring.

Using Fig. 1.2, determine the force that produces an extension of spring B that is 1.0 cm greater than the extension of spring A.

force =[1]

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0/N11/21/Q11(a)

- 1 A crankshaft is a shaped metal bar that is part of a car engine. It is free to rotate about an axis, as shown in Fig. 11.1.

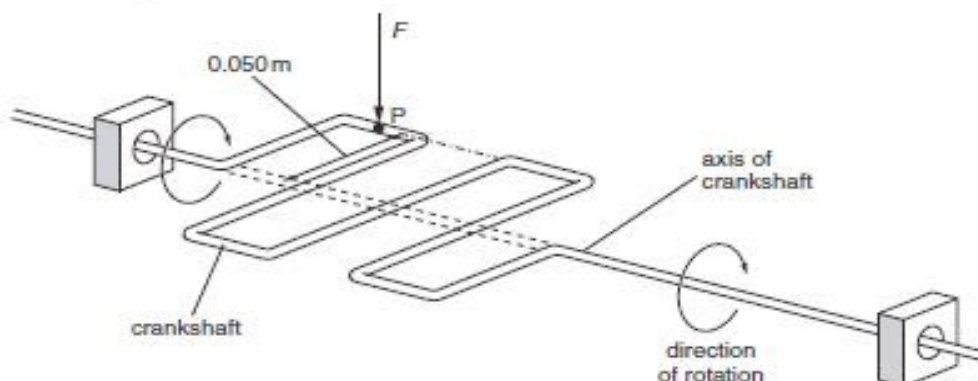


Fig. 11.1

When the crankshaft is horizontal, a vertical force F of 8200 N acts downwards on the crankshaft at P . This causes the crankshaft to rotate. The distance between P and the axis of the crankshaft is 0.050 m .

- (a) (i) State what is meant by the *moment* of a force.

.....
.....[2]

- (ii) The crankshaft is horizontal. Calculate the moment of F about the axis of the crankshaft.

moment =[2]

- (iii) The size and direction of the vertical force F , acting on the crankshaft at P , remain constant. Explain why the moment of F decreases as the crankshaft rotates through a small angle.

.....
.....[1]

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0/N11/22/Q1

A builder needs to determine the density of a solid cube of wood.

He places the 50 cm mark of a uniform metre rule on a pivot, so that the rule balances.

He then places the cube on the rule with its centre of gravity directly above the 75 cm mark. A mass of 0.050 kg is moved along the rule until balance is restored. This is shown in Fig. 1.1.

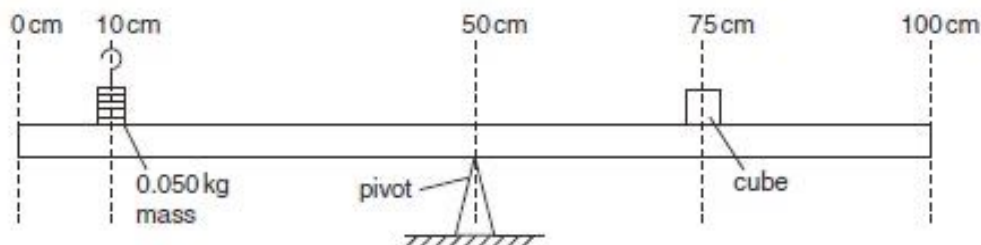


Fig. 1.1 (not to scale)

The rule is balanced when the 0.050 kg mass is at the 10 cm mark.

(a) Calculate the mass of the cube.

mass = [3]

(b) The cube has a volume of $1.6 \times 10^{-4} \text{ m}^3$. Determine the density of the wood.

density = [2]

Turning Effect and Deformation Questions P2 5054

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MARK SCHEME

O/N18/21/Q3

3(a) line continues from (7.0, 0.19) **and** curves upwards **B1**

3(b) (extension =) 8.5 (cm) **C1**

3.1–3.2 N **A1**

3(c) (limit of proportionality is) not reached **and** load shared / distributed equally **or** tension in each spring 4.5 N **or** limit of proportionality now 14 N

B1

or tension in each spring halved **or** tension in each spring < 7.0 N **B1**

M/J18/22/Q2

2(a) 150 and 220 in table **B1**

2(b) any use of proportionality **C1**

1.4 N **A1**

2(c)(i) 1st line larger at B **and** 2nd line larger at A **B1**

2(c)(ii) student loses energy / chemical energy decreases

or (gravitational) potential energy decreases **and** spring energy increases

B1

idea that increase in spring energy > change / decrease in (grav) PE (so student provides work needed) **B1**

O/N17/21/Q1

1(a) $\rho = m / V$ **or** $23 / (3.6 \cdot 0.35 \cdot 0.025)$ **or** $23 / 0.0315$ **C1**

730 kg / m³ **A1**

1(b)(i) 230 N **B1**

1(b)(ii) $\tau = Wx_{\perp r}$ **or** $230 \cdot 1.3$ **C1**

300 Nm **A1**

1(c) moment of painter / clockwise moment (about support) is greater than / not equal to / different from moment of

plank / anticlockwise moment

B1

O/N17/22/Q2

2(a) shape **B1**

size / volume / density **B1**

2(b)(i) 0.12 N **B1**

2(b)(ii) weight of first spring (increases extension) **or** weight of first spring (takes spring beyond the limit of proportionality) **B1**

2(b)(iii) where / point / load / extension / limit beyond which the extension is not directly proportional to load / extension-load graph is not straight

B1

2(c)(i) elastic potential energy **B1**

2(c)(ii) friction / air resistance mentioned **or** thermal energy / heat produced **B1**

M/J17/22/Q3

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3(a) force · distance

ignore force into distance

C1

force · **perpendicular** distance (from line of action to point / pivot) **A1**

3(b) any moment calculation seen, e.g. $F \cdot 22 = 80 \cdot 4$ **C1**

15 N **A1**

3(c) ($P =$) force / area **or** $80 / 0.0012$ **C1**

$6.7 \cdot 10^4$ Pa **A1**

O/N16/21/Q2

2 (a) 260 N **B1**

(b) (i) for a body in equilibrium **B1**

(total) clockwise moment = (total) anticlockwise moment **B1**

(ii) $F_1 d_1 = F_2 d_2$ **or** $260 \cdot 0.35$ **or** 91 **or** $F \cdot 0.65$ **C1**

$260 \cdot 0.35 = F \cdot 0.65$ **or** $260 \cdot 0.35 / 0.65$ **or** $91 = F \cdot 0.65$ **or** $91 / 0.65$ **C1**

140 N **A1** [6]

O/N16/22/Q3

3 (a) (i) 20 N **B1**

(ii) 1. ($\varphi =$) Fd **or** $20 \cdot 0.35$ **or** $20 \cdot 0.70$ **or** 14 **C1**

7.0 N m **A1**

2. friction (at hinge / seal) **or** air resistance **or** to cause an initial acceleration **B1**

(b) (for other directions) **perpendicular** distance is less **B1** [5]

O/N16/22/Q8

8 (a) (i) 0.83 – 0.86 N **B1**

(ii) line curved **B1**

line (curved) upwards **B1** [3]

(b) (i) ($P =$) $h\rho g$ **C1**

$0.035 \cdot 1000 \cdot 10$ **or** $3.5 \cdot 1000 \cdot 10$ **or** $35 \cdot 1000 \cdot 10$ **C1**

350 Pa **A1**

(ii) ($F =$) PA **or** $350 \cdot 0.0016$ **or** $350 \cdot 16$ **or** 5600 **C1**

0.56 N **A1**

(iii) 1.4 N **or** **(a)(i) + (b)(ii)** calculated **B1** [6]

(c) (i) (atmospheric pressure) exerts a downward force / pressure **B1**

(on top of the block)

(cancels out the) extra upward force / pressure **B1**

(ii) (vector) has direction (in addition to magnitude) **B1** [3]

(d) any **three** lines from

force due to water increases

force due to spring decreases

increased pressure (at base)

they add to give a constant value / weight of block **or** total force constant **B3** [3]

M/J16/21/Q2

(a) (i) limit of proportionality **B1**

(ii) 250 g **B1**

(iii) 2.5 N **B1**

(b) half the extension / 10 cm **B1**

each / both / another spring shares / distributes the weight / mass **B1**

or both springs bear / carry the load

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M/J16/22/Q3

3 (a) (i) C M1

(ii) data quoted to prove stretches more at end A1

or extensions / changes in length increase / are not the same (at higher loads)

(iii) 4.5 cm B1

(b) (tie rock to spring A) B1

• find weight / force / newtons using length or extension and graph or match readings (in table)

• find known weight / mass / force / N that gives same extension of spring

• use of proportionality with length or extension

• extension (in cm) / 1.6

(mass =) weight / g B1

or weight / gravitational field (strength)

O/N15/21/Q4

4 (a) (point) C immediately above tip of pivot (and in middle(vertically) of screwdriver (±1 mm)) B1

(b) (i) 0.64 N B1

(ii) arrow W vertically downwards through candidate's C or pivot B1

(c) no resultant force or upward force = downward force or force left = force right B1

no resultant moment (of force) or clockwise moment = anticlockwise moment B1 [5]

O/N15/22/Q2

(a) any two from different lines of:

distort / stretch / change in shape / squeezed / bends / deforms

compresses / change in size / volume / density / depth / height

change in temperature / gets hot(ter) / generates heat B2

(b) (i) straight line from origin B1

upward curve labelled / clear from limit of proportionality B1

(ii) permanent extension or spring is longer than it was originally B1 [5]

M/J15/21/Q2

2 (a) (i) force moves through a distance (in same direction) B1

(ii) chemical (potential) energy B1

(b) (i) 480 Nm B1

(ii) attempt to apply moments with two forces and distances C1

400 N A1

M/J15/22/Q1

(a) weight (pulls spring down and causes tension) B1

or force/pull of gravity

or mass is in gravitational field

(b) has a direction B1

(c) (i) $1\ l = l_0 + e$ or $l_0 = l - e$ or $e = l - l_0$ B1

2 36 cm B1

(ii) curve upwards after 10 N B1

O/N14/21/Q2

2 (a) (i) $F_1 \times d_1 = F_2 \times d_2$ or $(0.39 \times 0.40) / 0.30$ C1

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0.52 N A1

(ii) 0.052 kg or 52 g B1

(b) ($\rho =$) m / V or $52 / 60$ or $0.052 / 0.000\ 060$ or $0.052 / 60$ B1

$870 / 867 / 866.7$ kg / m^3 or 0.87 g / cm^3 or 8.7×10^{-4}

kg / cm^3 etc. B1 [5]

O/N14/22/Q1

1 (a) limit of proportionality (not breaking point) B1

(b) (i) 8.5 cm cao B1

(ii) 7.1 – 7.3 cm B1

(c) $4.0 \times (2.7 / 7.2)$ or $5.0 \times (2.7 / 9.0)$ or 1.5 (N) or read from graph or 11.2 (cm) C1

0.148 – 0.152 kg or 148 – 152 g A1 [5]

O/N14/22/Q2

2 (a) (i) Fd or 2.5×0.18 C1

0.45 N m A1

(ii) force not applied at right angles to the tap B1

(b) long(er) distance needs small(er) force (for same moment) or inversely related/proportional B1 [4]

O/N13/21/Q2

2 (a) $F_1x_1 = F_2x_2$ or $550 \times (0.86 \text{ or } 86)/(1.1 \text{ or } 110)$ C1

430 N A1

(b) both moments increase C1

girl's moment increases more or girl's moment > brother's
or anticlockwise moment greater

A1

see-saw tips down on girl's side B1 [5]

O/N13/22/Q9

9 (a) force $\times$ distance or $F \times d$ with F and d defined or $F \times d_{\text{perp}}$ C1

force $\times$ perpendicular distance or $F \times d_{\text{perp}}$ with F and d_{perp} defined A1 [2]

(b) (i) $1.6 \times 750 \times 1.2$ or 750×1.2 or 900 C1

5400 N m A1

2. mgh or $350 \times 10 \times 160$ or $350 \times 10 \times 1.6$ C1

$350 \times 10 \times 1.6$ or 5.6×10^5 C1

5600 J A1

(ii) friction at axle/boat or drag due to water
or chain lifted also B1

heat produced (ignore in sailors) or work done against friction/drag
or work done raising chain B1

(iii) same amount of work done or $P = E/t$ or $P = WD/t$ B1

in less time or power inversely proportional to time (ignore faster rate) B1 [9]

(c) clear/labelled diagram with ruler, fulcrum, at least two weights B1

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any three of the following points made in words:

balance ruler (on its own)

place weights on ruler so it balances

clockwise and anticlockwise moments equal or net moment zero

repeat (apply similar principles to other possible methods) B3 [4]

[Total: 15]

M/J13/22/Q2

2 (a) force $\times$ distance M1

perpendicular distance A1

(b) (i) $T \times 8$ or 2000×2 seen C1

500 N A1

(ii) (two forces) equal (in magnitude) B1

(two forces) opposite (in direction) B1 [6]

M/J13/22/Q3

(a) (i) $(W=) Fd$ C1

or 90×0.3

or 90×30

27 J A1

(ii) $(P=) W/t$ C1

or Fd/t

or $27(\times 20)/60$

or $27/3$

9(0) W A1

(b) (i) $800 \times 30/180$ C1

or $800/6$

or 6 seen

or proportionality clearly used

133 or 130 cm A1

(ii) extension more than 143 cm or (extra) extension > 10 cm B1 [7]

or (some) extension permanent

O/N12/22/Q1

(a) appropriate apparatus e.g. ruler, weights, fulcrum B1

action e.g. balance weights on each side B1

one of: force/mass $\times$ distance or calculate moment B1

vary or repeat B1

(b) $F \times d$ or 8.0×0.15 C1

1.2 N m (not J) A1 [6]

O/N12/22/Q2

2 (a) (i) 4.5 kg B1

(ii) axes labelled with quantity and unit B1

linear scale B1

straight line from clear (0,0) to correct point B1

(b) answer from candidate's line B1 [5]

O/N11/21/Q11(a)

11 (a) (i) force $\times$ distance C1

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force $\times$ perpendicular distance (from the axis) A1

(ii) 8200×0.05 C1

410 N m A1

(iii) (perpendicular) distance reduced/force not perpendicular/only a component of the force is perpendicular B1

O/N11/22/Q1

(a) $m_1(g)x_1$ or $m_2(g)x_2$ or 2 or one of these in numbers or 40 and 25 seen C1

$0.050 \times (10) \times 40 = m_2 \times (10) \times 25$

or anticlockwise moment = clockwise moment C1

0.080 kg or 80 g A1

(b) $(\rho / d =) m/V$ or $0.08/1.6 \times 10^{-4}$ C1

500 kg / m³ or 0.50 g / cm³ A1 [5]